Universidade Estadual de Mato Grosso do Sul - UEMS
2º ano de Ciência da Computação
Avaliação Substitutiva de Linguagens Formais e Autômatos - 02/12/2024
Professor Osvaldo Vargas Jaques

ALUNO:_____ **RGM:**_____

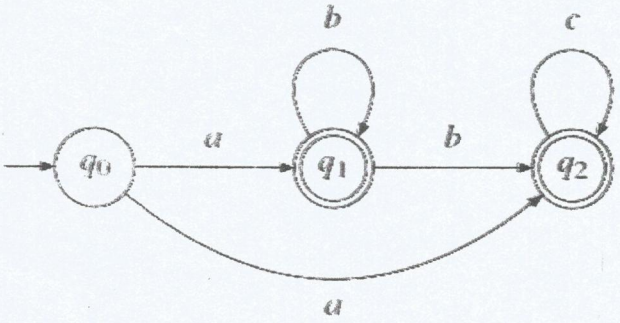
□

1. (2,5 pontos) Dada a gramática abaixo, esboce o diagrama de estados e a descrição formal do AF equivalente:

AF ≠ AP

$$A \rightarrow 0A \mid 0B \mid 1A \mid 1C$$
$$B \rightarrow \epsilon$$
$$C \rightarrow 0A$$

2. (2,5 pontos) Dado o **AFN** abaixo, encontre a expressão regular equivalente.



3. (2,5 pontos) Dada a expressão regular $aa(ab)^*(a \bigcup b)^*(ba)^*$, encontre o AF.

4. (2,5 pontos) Dada a gramática abaixo:

$$S \rightarrow AB \mid C$$
$$A \rightarrow aAb \mid ab$$
$$B \rightarrow cBd \mid cd$$
$$C \rightarrow aCd \mid aDd$$
$$D \rightarrow bDc \mid bc$$

Utilizando a string *aabbccdd*, verifique se a gramática é ambígua. Se for, **mostre as duas**:

a) Utilizando a string *aabbccdd* construa a árvore sintática.

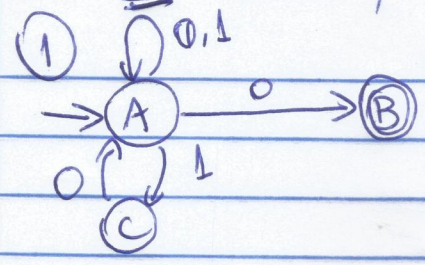
b) Essa gramática é ambigua? Responda por quê.

c) Dê a descrição formal da gramática

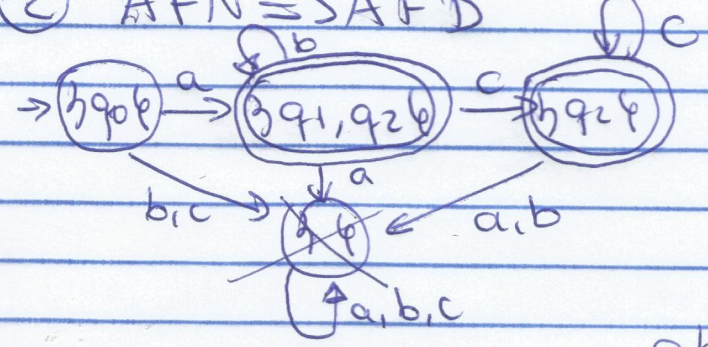
d) Encontre o autômato de pilha (AP) equivalente e dê a descrição formal desse AP

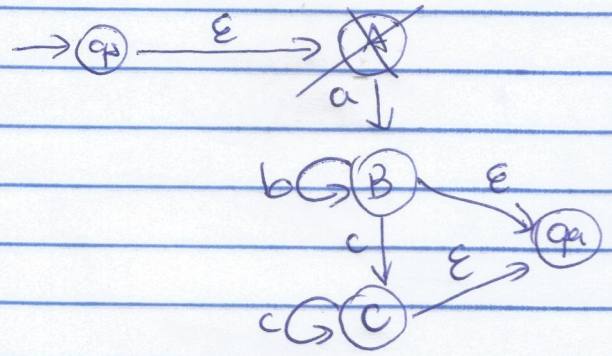
"Ninguém, depois de acender uma candeia, a põe em lugar escondido, nem debaixo do alquire, mas no velador, a fim de que os que entram vejam a luz."Lucas 11:33
De nada vale o maior conhecimento que uma pessoa possa ter se ela não passá-lo adiante."

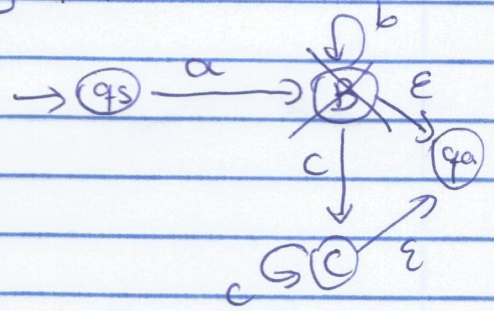
AF de GR, AF ≠ AP

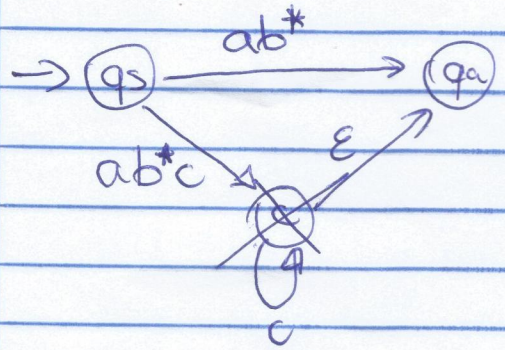
① 

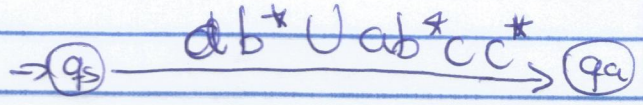
② AFN => AFD











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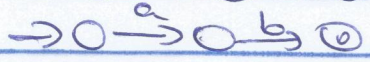
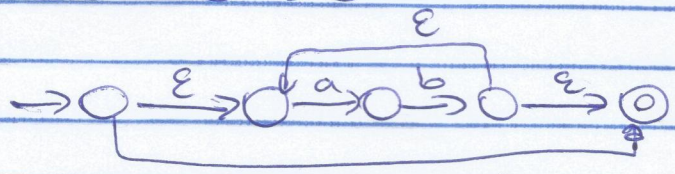
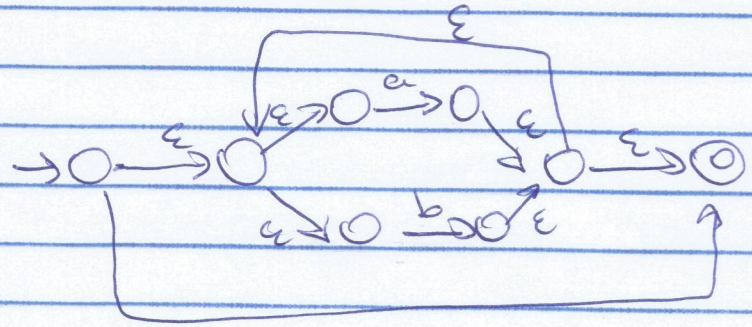
③

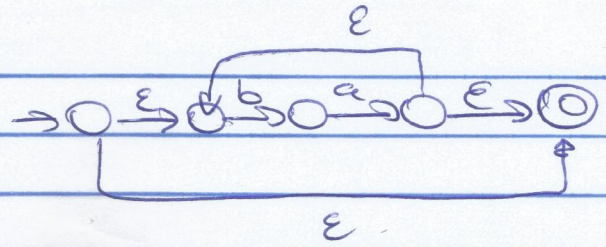
aa

ab 

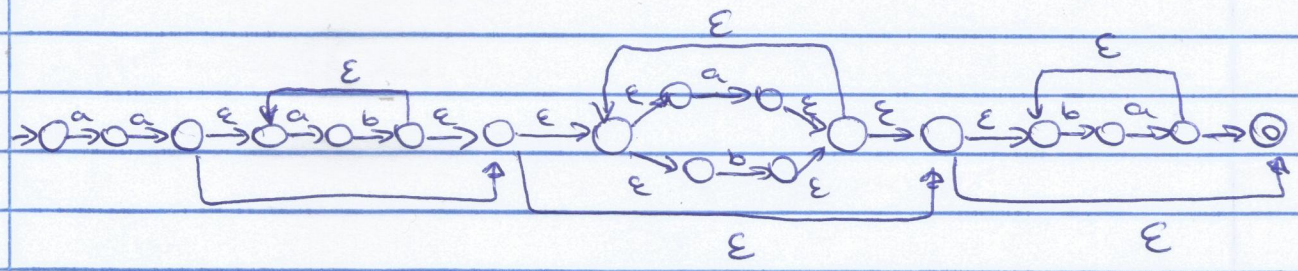
$(ab)^*$ 

$(a \cup b)^*$ 

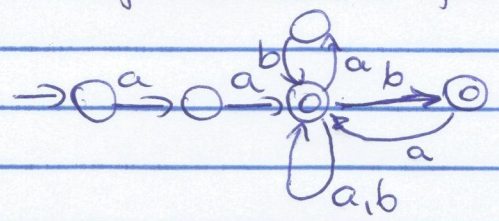
$(ba)^*$



Assim $aa(ab)^*(a \cup b)^*(ba)^*$



Após experiência, é fácil ver



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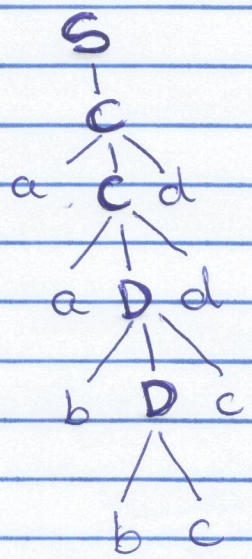
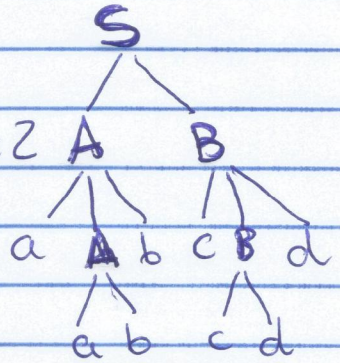
(4)

a)



Arvore 1



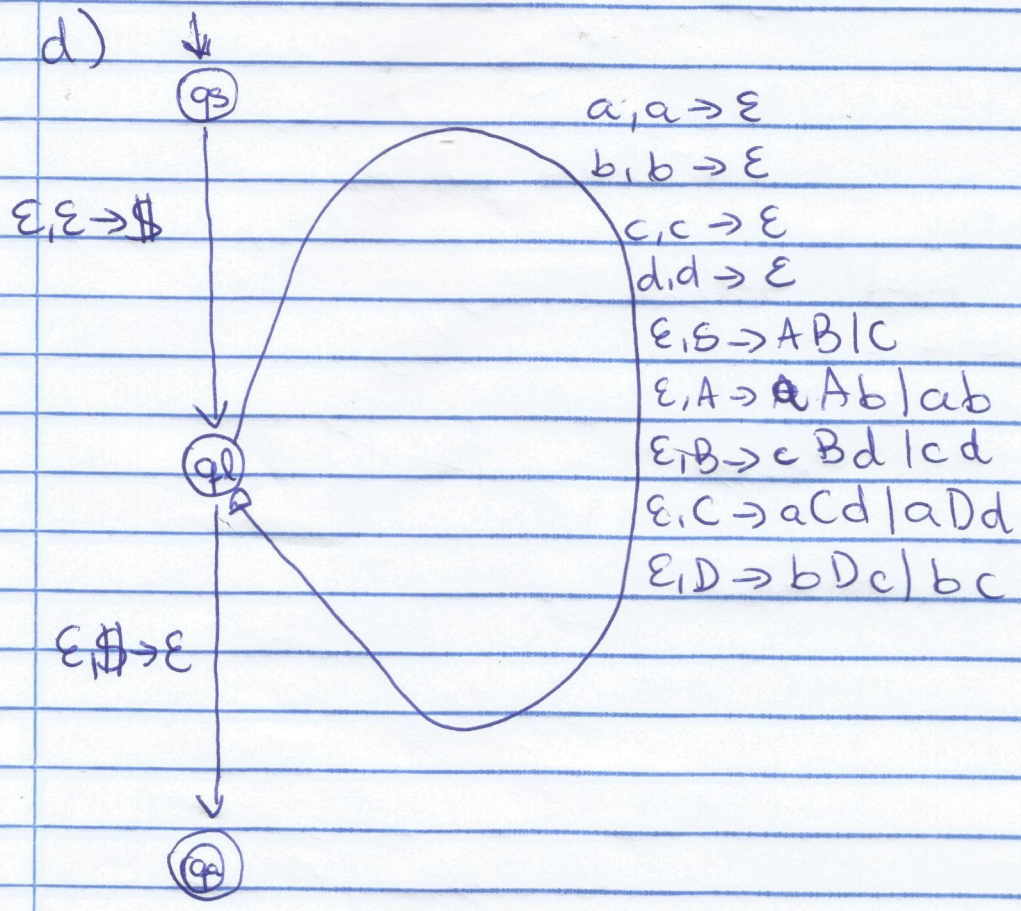
Arvore 2

b) Como temos mais de uma árvore sintática em a), a gramática é AMBÍGUA.

c) $G = (V, T, P, S)$

$G = (\{S, A, B, C, D\}, \{a, b, c, d\}, \{S \to AB \mid C,$
$A \to aAb \mid ab,\ B \to cBd \mid cd,\ C \to aCd \mid aDd,$
$D \to bDc \mid bc\}, S)$

d)



Definição formal:

$P = (Q, \Sigma, \Gamma, \delta, q_0, F)$

onde:

Γ: alfabeto da pilha

$\delta : Q \times \Sigma_\varepsilon \times \Gamma_\varepsilon \to \mathcal{P}(Q \times \Gamma_\varepsilon)$

$\mathcal{P}$: conjunto das partes

$P = (\{q_s, q_l, q_a\}, \{a, b, c, d\}, \{a, b, c, d, \$, A, B, C, D, S\},$
$\delta, q_s, \{q_a\})$

onde
δ:

$\delta(q_s, \varepsilon, \varepsilon) = \{(q_l, \$)\}$
$\delta(q_l, a, a) = \{(q_l, \varepsilon)\}$
$\delta(q_l, b, b) = \{(q_l, \varepsilon)\}$
$\delta(q_l, c, c) = \{(q_l, \varepsilon)\}$
$\delta(q_l, d, d) = \{(q_l, \varepsilon)\}$
$\delta(q_l, \varepsilon, S) = \{(q_l, AB), (q_l, C)\}$
$\delta(q_l, \varepsilon, A) = \{(q_l, aAb), (q_l, ab)\}$
$\delta(q_l, \varepsilon, B) = \{(q_l, cBd), (q_l, cd)\}$
$\delta(q_l, \varepsilon, C) = \{(q_l, aCd), (q_l, ad)\}$
$\delta(q_l, \varepsilon, D) = \{(q_l, bDc), (q_l, bc)\}$
$\delta(q_l, \varepsilon, \$) = \{(q_a, \varepsilon)\}$